La Scuola d'Italia / Dr. Huson / IB Math 11
13 October 2026

Nickname:

1.11 Classwork: Geometric Series

1. The sum of the first n terms of a geometric sequence is

$$S_n = \frac{u_1\,(r^n - 1)}{r - 1} \qquad \text{or} \qquad S_n = \frac{u_1\,(1 - r^n)}{1 - r}, \qquad r \neq 1.$$

Consider the geometric series

$$2 + 6 + 18 + 54 + \cdots$$

(a) Find S_5, the sum of the first five terms, by writing out the five terms and adding them.

(b) Use the first formula above to find S_5. Show the values you substitute.

(c) Find S_{10}, the sum of the first ten terms.

(d) The sum of the first n terms is 177 146. Find n.

Nickname:

2. Two college graduates, Luca and Sofia, start their first jobs. Their starting annual salaries are close to the average for college graduates where they live. Luca works in New York City with a starting annual salary of \$70 000. Sofia works in Italy with a starting annual salary of €30 000. Each of their annual salaries increases by 6% every year.

   (a) Find Luca's annual salary in his third year.

   (b) Find the total amount Luca earns in his first three years.

   (c) Find the number of complete years Luca must work for his total earnings to first exceed \$1 000 000.

   (d) Find the number of complete years Sofia must work for her total earnings to first exceed €1 000 000.